

phase : project design phase

Date	29 september 2026
Team ID	SWTID-2026-9366
project name	Script controlled ACL
Maximum marks	4

The Project Design Phase bridges the gap between initial requirements gathering and actual implementation/development. It translates system goals into detailed architectural, functional, and technical blueprints.

1. KeyObjectives

- Define System Architecture: Establish high-level infrastructure, software design patterns, and data integration models.
- Detail Functional & Technical Requirements: Map out exact workflows, data models, UI wireframes, and business rules.
- Mitigate Technical Risk: Identify dependencies, security constraints, and performance bottlenecks prior to build.
- Establish Baselines: Lock down scope, design specifications, and test criteria for development teams.

2. Core Activities & Deliverables

Category	Key Activities	Output Deliverables
Architectural Design	System boundary planning, component modeling, technical stack selection, integration protocols (APIs).	System Architecture Document (SAD), Integration Maps
Data & Schema Design	Entity relationship mapping, database schema definition, field-level access controls, data dictionary.	ER Diagrams, Data Dictionary, Security Matrix

UI/UX Design	Userjourneymapping, wireframing, high-fidelity mockups, design system alignment.	Wireframes,Interactive Prototypes, UI Specs
Functional & Technical Logic	Processflowmapping,business logic definition, validation rules, user story writing.	TechnicalDesignDocument (TDD), Detailed Specs
Testing & Quality Strategy	Test case scenario definition, acceptance criteria mapping, performance benchmarking setup.	TestPlan,TraceabilityMatrix

3. High-Level Process Workflow

1. Input:BusinessRequirementsDocument(BRD)&StakeholderVision
2. ConceptualDesign:Formulatehigh-levelarchitectureoptionsandevaluatetrade-offs.
3. DetailedDesign:Draftgranularsystemmodels,databasetables,componentstructures,anduser flows.
4. Review&Validation:Conductarchitecturereviews,securityassessments,andUX walkthroughs.

5. Output&Sign-off:FinalizeblueprintsandhandofftodevelopmentandQAteams.

4. Key Considerations & Guardrails

- Security & Compliance: Define role-based access control (RBAC), encryption methods, and compliance standards early.
- Scalability: Plan infrastructure capacity and performance targets to handle future load.
- Maintainability: Ensure code standards, modular designs, and documentation lower long-term technical debt.

Would you like to focus on a specific aspect of this phase, such as technical architecture templates, UI/UX workflows, or security design?